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Editors
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Dear Editors,

Please consider the manuscript

*Differential Unrooting and Stable Incident-Edge Marginals
in Weighted Random Forests*

by Shigang Ou for publication as a research article.

The general negative-correlation problem for weighted random forests remains open. The manuscript proves a local result of a stronger polynomial kind: for every vertex of every finite loopless multigraph, and for arbitrary nonnegative inhomogeneous edge activities, the complete incident-edge marginal has a multiaffine real-stable generating polynomial. The conclusion persists after feasible coordinate conditioning on edges outside the star. It implies conditional negative association within a star, negative correlation for all adjacent edge pairs without a common-activity or asymptotic assumption, and a Poisson-binomial law for every star count.

The proof is based on a component-marked forest polynomial and an exact “differential unrooting” identity that converts the rooted matrix-forest determinant into the required unrooted polynomial by stability-preserving second-derivative operators. The paper includes a detailed coefficientwise proof audit. An accompanying finite symbolic regression test is diagnostic only; no theorem depends on computation.

The manuscript explicitly does not claim negative correlation for disjoint edges or global strong Rayleighness. It is original, self-contained, and is not submitted to another journal. Automated tools assisted proof exploration, finite regression checks, literature triage, and language editing; this assistance is disclosed in the manuscript, and the author accepts responsibility for all proofs, citations, and text.

Thank you for your consideration.

Sincerely,

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of Sciences
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